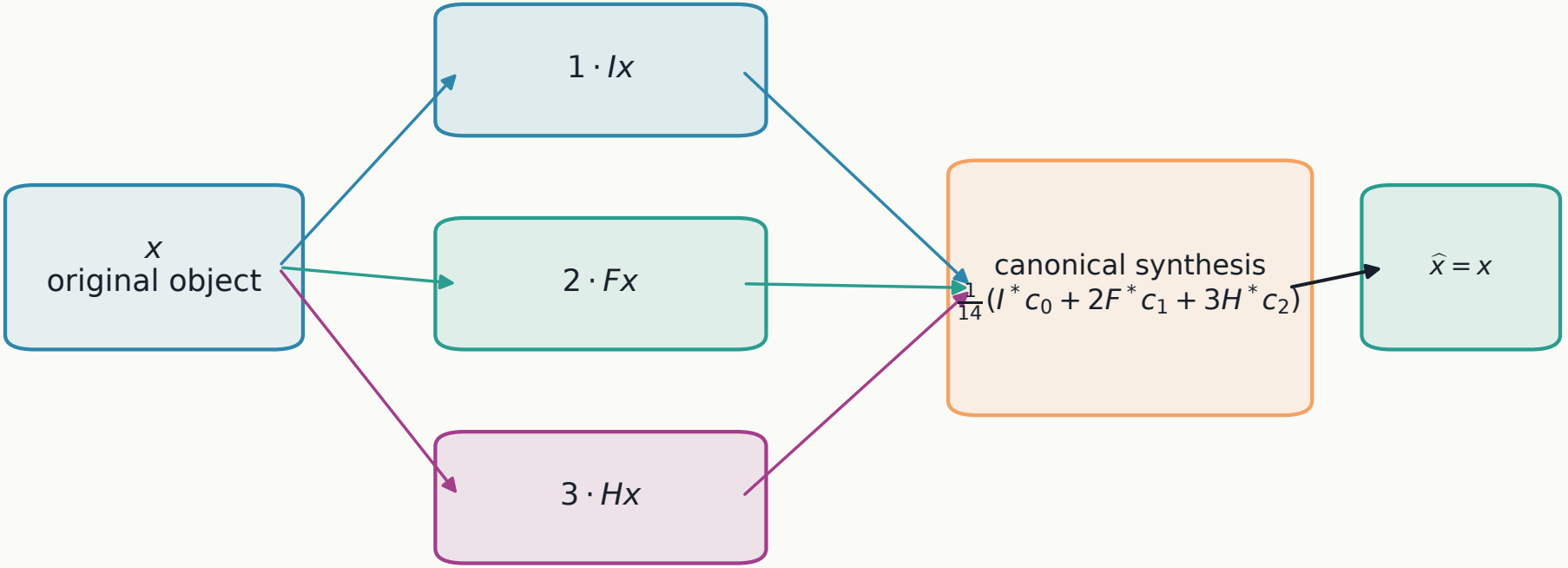
The weighted spatial-Fourier-Haar portfolio is a finite tight frame



$$SA = (1^2 + 2^2 + 3^2)^{-1}(1^2 I + 2^2 I + 3^2 I) = I.$$